The Gravity–Light Interaction Survives Cyclic Causal Reduction through ℓ_3

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Working draft, July 2026

Abstract

We transfer the classical gravity–Maxwell BV interaction on a compact positive Berger-clock background from a complete 64-row presentation to a typed cyclic 36-row retained complex. The contraction carries the exact odd pairing and the unary causal Green data. In the repository's suspended graded-symmetric convention the transferred binary operation has 1,474 coefficients and the mixed ternary operation has 25,950 coefficients. The latter is not killed by reduction: it contains 7,614 gravity-output terms with two Maxwell inputs and 18,336 Maxwell-output terms with one Maxwell input.

The homological exchange channel is computed rather than omitted. It contains 342 canonical full-complex coefficients in the only nonempty channel, all supported on full row 38, and the certified retained projection annihilates that row. A one-entry projection mutation exposes all 342 terms. Thus the retained exchange contribution vanishes for a structural reason while the direct q_3 contact survives. All 36 arity-three L_∞ rows close. Lowering ℓ_3 with the retained typed pairing independently reproduces all 25,662 physical quartic coefficients with zero cyclicity defects; changing the Maxwell pairing weight from two to one produces 17,108 exact defects on 14 rows.

Together with the cyclic causal Cartan contraction for the background-fixing helical generator K_{Berger}, these calculations establish

$$\boxed{\text{The gravity–light interaction survives cyclic causal reduction through } \ell_3.}$$

The statement is classical and carries separate LOCAL-ALGEBRAIC and LORENTZIAN-CAUSAL dependencies. The causal claim concerns the unary Green contractions and the two-sided-causal-hull Cartan primitives; the Taylor operations themselves are support-local. The retained complex is not yet the Einstein-like/extra-Weyl residual model. Its mixing table, the 288-coefficient ghost/antifield cyclicity replay, Hadamard products, QME restoration, and every quantum or particle interpretation remain open.

AI-use disclosure and computational accountability

OpenAI GPT-5 was used substantively for mathematical synthesis, verification code, claim auditing, and manuscript drafting. The human author directed the programme and remains responsible for the claims, citations, and final text. Every numerical count in the theorem is an exact sparse-coefficient count over $\mathbb{Q}(\sqrt{10})$ and is tied to a deterministic certificate. No floating point calculation enters an identity, rank, cyclicity test, or mutation witness. Submission remains conditional on final human review.

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Contents

1 Question and answer

The free BV complex identifies which classes survive gauge reduction, but it does not determine whether nonlinear gravity–matter vertices survive the same reduction. A transferred interaction can disappear in three distinct ways: its direct contact term can project to zero; homological exchange terms can cancel it; or cyclicity and the L_∞ identities can fail after compression. Conversely, a nonzero tensor on a gauge-fixed carrier need not descend to any physical branch until its residual projection is computed.

This paper answers the algebraic question through ternary order for the gravity–Maxwell extension of the positive Berger-clock complex. Write

$$\boxed{\text{The gravity–light interaction survives cyclic causal reduction through } \ell_3.} \tag{1}$$

Here "survives" means that the explicitly transferred mixed ternary bracket is nonzero and obeys the retained arity-three identity. "Cyclic" means both that the contraction is typed cyclic and that the complete physical quartic form obtained by lowering ℓ_3 is independently invariant under cyclic transposition. "Causal reduction" means that the unary contraction imports the advanced/retarded Green homotopies and that the compatible K_{Berger} Cartan primitives through arity three have support in the two-sided causal hull. It does *not* mean that a time-ordered interacting quantum theory has been constructed.

The theorem stops one rung before physical branch interpretation. The dynamical residual carrier must still be split into Einstein-like, extra-Weyl, and Maxwell branches. Separately, the even Weyl-square and odd Pontryagin directions form a deformation/vertex basis; the odd topological class is not a third particle branch. This category separation is built into the receiving contract for the next calculation.

2 Complexes, conventions, and claim boundary

Let $(\mathcal{C}_{64}, q_1, \omega_{64})$ be the complete typed classical gravity–clock–Maxwell BV complex at the frozen rational Berger fixture. Its degree multiplicities are

$$6, 26, 26, 6 \qquad \text{in degrees } -1, 0, 1, 2. \tag{2}$$

Let $(\mathcal{R}_{36}, \ell_1, \omega_{36})$ denote the retained typed carrier, with degree multiplicities

$$4, 14, 14, 4. \tag{3}$$

The exact contraction is

$$(\mathcal{C}_{64}, q_1) \underset{\iota}{\overset{\pi}{\rightleftarrows}} (\mathcal{R}_{36}, \ell_1), \qquad \pi\iota = \mathrm{id}, \qquad q_1 S + S q_1 = \mathrm{id} - \iota\pi, \tag{4}$$

with side conditions and typed cyclic adjoint relations fixed by the carrier certificate. The coefficients lie in $\mathbb{Q}(\sqrt{10})$. The Maxwell field–equation pairing has absolute weight two, while the gravity weight is one. The signed odd-pairing ledger contains ten gravity entries of sign -1 and four Maxwell entries of sign $+2$ on the physical field–equation pairs.

The based classical BV vector field is known through cubic Taylor order,

$$q = q_1 + q_2 + q_3 + O(4), \tag{5}$$

in the repository's suspended graded-symmetric factorial convention. The full tensors are action-derived, support-local, typed cyclic, and equivariant under the background-fixing helical generator

$$K_{\mathrm{Berger}} = D - \omega R. \tag{6}$$

Raw cylinder translation D is affine at the rotating background and is not substituted for K_{Berger}.

The result combines two dependency layers:

Layer	Certified content	Explicit limitation
LOCAL-ALGEBRAIC	exact q_2, q_3, ℓ_2, ℓ_3, identities and cyclicity	no residual branch interpretation
LORENTZIAN-CAUSAL	unary Green homotopies and K-Cartan support through arity three	no Hadamard products or QME

Neither layer is silently promoted to RESIDUAL-TRANSFERRED or LORENTZIAN-CERTIFIED quantum status.

3 Ternary homological transfer

The first two transferred Taylor operations are

$$\ell_1 = \pi q_1 \iota, \tag{7}$$
$$\ell_2 = \pi q_2(\iota, \iota). \tag{8}$$

Define the second Taylor component of the inclusion by

$$I_2 = -S q_2(\iota, \iota). \tag{9}$$

Then the ternary transfer formula is

$$\ell_3 = \pi q_3(\iota, \iota, \iota) + \sum_{\sigma \in \mathrm{Sh}(2,1)} \epsilon(\sigma)\, \pi q_2\big(I_2(\iota_{\sigma(1)}, \iota_{\sigma(2)}), \iota_{\sigma(3)}\big). \tag{10}$$

The first term is the contact contribution. The second line is the exchange contribution $q_2 S q_2$.

Lemma 3.1 (Exact exchange support)**.** *For the mixed gravity–Maxwell ternary operation, the gravity-outer/mixed-inner exchange channel contains* 144 *outer/inner coefficient pairs. The three graded unshuffles produce* 324 *signed contributions and* 342 *canonical full-complex PBW coefficients. Every one has full output row* 38*. The retained projection has no support on that row, so this channel vanishes after projection. The mixed-outer/gravity-inner and mixed-outer/mixed-inner channels have no outer/inner pairs.*

Proof. The exact consumer reconstructs the 96 gravity and 12 mixed coefficients of I_2, including their full-row supports, before composing them with q_2. PBW normalization gives the stated counts. Direct application of π annihilates the 342 row-38 coefficients. As a negative control, adding the single mutant entry

$$\pi_{\mathrm{mut}}(\text{retained row } 0, \text{full row } 38) = 1 \tag{11}$$

exposes all 342 coefficients. Thus the zero is caused by the certified projection and is not an empty table hard-coded by the consumer. □

Corollary 3.2 (Contact form of the retained mixed bracket)**.** *The retained mixed exchange contribution is zero and*

$$\ell_3^{\mathrm{mixed}} = \pi q_3^{\mathrm{mixed}}(\iota, \iota, \iota). \tag{12}$$

This equality is a computed property of the declared contraction, not a general truncation of the homotopy-transfer formula.

4 The algebraic survival theorem

Theorem 4.1 (Gravity–light survival through ℓ_3)**.** *On the frozen positive Berger-clock gravity–Maxwell BV complex, the typed cyclic contraction* (4) *transfers the action-derived Taylor data through ternary order with the following exact properties.*

1. *The retained mixed binary operation ℓ_2 contains* 1,474 *canonical coefficients.*

2. *The retained mixed ternary operation ℓ_3 contains* 25,950 *canonical coefficients on* 18 *nonzero output rows.*

3. *All retained exchange sectors vanish for the support reason in Lemma 3.1; the nonzero ternary operation is the transferred contact term.*

4. *The retained arity-three identity closes on all* 36 *rows:*

$$[\ell_1, \ell_3] + \frac{1}{2}[\ell_2, \ell_2] = 0. \tag{13}$$

5. *The mixed bracket has* 7,614 *gravity-output coefficients with two Maxwell inputs and* 18,336 *Maxwell-output coefficients with one Maxwell input. Hence it is nonzero in both gravity–light directions.*

A localized one-coefficient mutation of ℓ_3 creates two exact arity-three defects.

Proof. Equations (8)–(10) are evaluated in exact sparse PBW normal form over $\mathbb{Q}(\sqrt{10})$. The independent consumer parses all 59,598 coefficients of the full mixed q_3 payload and reproduces all 25,950 retained contact coefficients with no missing, extra, or changed terms. Lemma 3.1 evaluates every possible mixed exchange channel. Direct coefficient comparison of (13) gives zero defects on all rows. The input-sector and output-sector ledgers give the two nonzero counts in item 5. Finally, the mutation changes one retained coefficient and produces two normalized defects, preventing acceptance by a consumer that merely assumes the expected answer. $\square$

The theorem is stronger than the statement that Maxwell fields occur in the full action. It says that their ternary coupling remains present after the explicit gauge-fixed-to-retained homological reduction and after every arity-three identity is imposed. It is weaker than a residual particle vertex: $\mathcal{R}_{36}$ is retained, not yet minimal or decomposed into its dynamical branches.

5 Physical quartic cyclicity

Lowering the output of ℓ_3 with the odd pairing gives a four-linear functional. For physical degree-zero inputs, cyclic transposition of the first input requires formal adjunction of every PBW derivative through the invariant volume form. If w is a derivative word, the exact formal adjoint distributes its letters over the other three slots with sign $(-1)^{|w|}$ and the corresponding multinomial multiplicities.

Proposition 5.1 (Independent physical quartic cyclicity)**.** *Every degree-zero physical coefficient of the lowered retained ℓ_3 satisfies exact cyclic transposition. There are*

$$25{,}662 = 7{,}506_{\text{gravity output}} + 18{,}156_{\text{Maxwell output}} \tag{14}$$

such coefficients, and the defect count is zero on every row. Replacing the Maxwell field–equation weight two by weight one produces 17,108 *defects on* 14 *rows.*

Proof. The verifier constructs both sides independently from the retained tensor, the typed pairing, and exact PBW integration by parts. The signed pairing orientations are retained as a separate convention ledger; only their absolute component multiplicities enter the physical field–equation transpose. Equality holds coefficientwise. The weight mutation changes no coefficient of ℓ_3 itself and therefore tests the lowering convention rather than the imported interaction. $\square$

The remaining

$$25{,}950 - 25{,}662 = 288 \tag{15}$$

coefficients are the ghost/antifield completion. Their cyclicity is part of the classical typed-transfer theorem but has not yet received the same independent transpose replay. Proposition 5.1 is therefore a complete physical-quartic statement and explicitly not an independently replayed full-BV cyclicity theorem.

6 Causal compatibility through ternary order

The causal part of the headline is not inferred from finite PBW tables. It is supplied by the independently certified unary Green chain contractions and the coupled cyclic Cartan theorem. Let $\Lambda_{64,\pm}$ denote the same-sided advanced and retarded unary homotopies. Their chain identities, causal supports, and formal-adjoint reversal are imported on the complete carrier. The local Taylor operations and the contraction intertwine K_{Berger}.

Let $\iota_K^{(1)}$ be the unary Cartan primitive. The higher sources are

$$A_K^{(2)} = [q_2, \iota_K^{(1)}] - L_K^{(2)}, \tag{16}$$

$$A_K^{(3)} = [q_3, \iota_K^{(1)}] + [q_2, \iota_K^{(2)}] - L_K^{(3)}. \tag{17}$$

At the frozen background $L_K^{(2)} = L_K^{(3)} = 0$. The L_∞ identities, graded Jacobi identity, and K-equivariance give

$$[q_1, A_K^{(2)}] = [q_1, A_K^{(3)}] = 0. \tag{18}$$

Raw causal primitives are cyclicized with the exact Reynolds projectors

$$\mathrm{Cyc}_3 = \frac{1+\tau+\tau^2}{3}, \qquad \mathrm{Cyc}_4 = \frac{1+\tau+\tau^2+\tau^3}{4}. \tag{19}$$

Theorem 6.1 (Cyclic causal Cartan compatibility)**.** *The coupled* 64*-row complex and its typed* 36*-row retained contraction obey the* K_{Berger} *Cartan identities through arity three. The higher cyclic primitives satisfy*

$$\operatorname{supp}\big(\iota_K^{(n)}(f_1,\dots,f_n)\big) \subseteq J\left(\bigcup_i \operatorname{supp} f_i\right), \qquad n \le 3. \tag{20}$$

The Taylor operations are support-local and no spatial inverse or mode projector enters the construction.

The higher cyclic primitives use the two-sided causal hull. They are not claimed to be separately retarded or advanced; the unary homotopies retain that stronger one-sided property. Theorem 6.1 establishes compatibility of the surviving interaction with the causal reduction. It does not construct interacting retarded products, a Hadamard state, or a Lorentzian quantum master equation.

7 The residual mixing table

The next calculation determines what the surviving tensor means physically. It requires a committed, content-addressed residual branch manifest rather than an inferred basis. The dynamical carrier must contain

$$\mathcal{R}_{\mathrm{dyn}} = \mathcal{R}_{\mathrm{Einstein}} \oplus \mathcal{R}_{\mathrm{extra\ Weyl}} \oplus \mathcal{R}_{\mathrm{Maxwell}}, \tag{21}$$

together with exact inclusion, projection, pairing, parity, real structure, and K_{Berger} weights. The mixing entries are then

$$M^a{}_{bcd} = \pi^a_{\mathrm{dyn}}\, \ell_3\big(\iota_b^{\mathrm{dyn}}, \iota_c^{\mathrm{dyn}}, \iota_d^{\mathrm{dyn}}\big). \tag{22}$$

Every claimed zero must carry an exact support or cancellation witness, and graded symmetry and cyclicity must be replayed in branch coordinates.

The deformation/vertex basis is separate:

$$e_{C^2} = \frac{[W_+^2] + [W_-^2]}{\sqrt{2}}, \qquad o_{C\widetilde{C}} = \frac{[W_+^2] - [W_-^2]}{\sqrt{2}}. \tag{23}$$

Its Euler–Lagrange map has the required rank pattern

$$\operatorname{rank} E(e_{C^2}) = 1, \qquad E(o_{C\widetilde{C}}) = 0, \tag{24}$$

with a separate Pontryagin transgression witness. The odd topological direction can be central or inert as a deformation class without becoming a third dynamical particle branch.

At the time of this draft the strict projection consumer is ready, but the authoritative branch-basis manifest has not been supplied. Accordingly this section freezes the question and the formula, not its answer. The algebraic survival theorem does not depend on that input.

8 Interpretation and nonclaims

The result provides a precise first meaning of "additional physics beyond Einstein" at nonlinear order: the retained classical interaction contains both a gravitational response to two Maxwell inputs and a Maxwell response to a gravity–Maxwell pair. These channels survive exact cyclic transfer and the ternary homotopy identity. Once the branch manifest lands, Equation (22) will decide whether the interaction closes on the Einstein-like branch, couples it to the extra-Weyl branch, or acts only through the deformation directions.

Several stronger conclusions do not follow.

1. The bracket changes no unary kinetic operator. It therefore introduces no new negative kinetic direction at this order, but this is not a unitarity or positive-Hilbert-space theorem.
2. A nonzero retained bracket is not yet a photon or graviton scattering amplitude. Residual projection, causal state selection, and asymptotic or relational observables are still required.
3. The odd Pontryagin direction is a topological deformation class, not a one-particle state.
4. The vanishing one-particle cohomology of the separate free conformal cylinder problem is not converted here into an interacting vacuum theorem.
5. No loop coefficient, anomaly cancellation, QME restoration, Hadamard state, renormalized product, or quantum interaction has been constructed.
6. No arity-four or convergent all-orders L_∞ or Cartan theorem is claimed.

The natural provisional interpretation is therefore a deformation/vertex theory rather than a particle theory. The mixing table is precisely the next test of how much of that statement remains true after residual descent.

9 Reproducibility and certificate map

The algebraic theorem is pinned to the classical hardening commit `e99d0c1d39490de5261fc6ca1dc2aeaa0d149655`. Its principal artifacts are

Artifact	Role
`BERGER_COUPLED_K_CARTAN_THROUGH_ARITY_THREE`	full/retained cyclic causal K-Cartan theorem
`BERGER_RETAINED_MIXED_ELL3_INDEPENDENT_ACCEPTANCE`	exact independent $64 \to 36$ transfer and mutation witnesses
`BERGER_RETAINED_MIXED_ELL3_PHYSICAL_CYCLICITY`	independent physical quartic cyclicity and pairing-weight mutation
`BERGER_RESIDUAL_MIXED_ELL3_BRANCH_PROJECTION_READINESS`	fail-closed input contract for the mixing table

The scoped reproduction commands are

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_coupled_k_cartan_through_arity_three.py --check --guards
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_coupled_k_cartan_through_arity_three.py
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_acceptance_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_acceptance
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_physical_cyclicity_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_physical_cyclicity
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_residual_ell3_branch_projection_readiness
```

The companion machine-readable claim map records the dependency hashes, positive claims, explicit nonclaims, and the fact that this manuscript is a working draft rather than a theorem freeze.

10 Outlook

Paper 11 can now develop on two rails. The first is editorial: sharpen the abstract transfer theorem, extract compact coefficient tables to a supplement, and obtain specialist review of the cyclic-sign and causal-support conventions. The second is mathematical: import the residual branch basis and compute Equation (22). A nonzero Einstein/extra-Weyl entry would turn the additional Weyl physics into an explicit interaction channel; a block-diagonal table would establish nonlinear closure of the Einstein-like direction through this order; and a central topological action would isolate the vertex theory from the dynamical carrier.

The algebraic theorem does not need to wait for that verdict. What the mixing table changes is its physical interpretation, not the existence of the surviving cyclic gravity–light bracket.

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